

LLM 4-Days Report

Xisoblashlar

③ Parametrlar

$$12 \cdot L \cdot d^2$$

④ Xotira sog'adimi?

$$16B \times N$$

⑤ Compute Budgeti

$$6 \cdot N \cdot D$$

⑥ Lagit (wall-clock)

$$6ND / (\text{GPU} \cdot \text{peak} \cdot \text{MFU})$$

⑦ Nux

$$\text{soat} \cdot \text{GPU} \cdot \$/\text{soat}$$

⑧ Samaradorlik

$$\text{MFU} \uparrow, \text{BF} \neq 6, \text{checkpoint}$$

Frontier Modellari - hozirgi paytdagi eng kuchli modellari.
ChatGPT, Claude, Gemini va boshqalar.

Embeddings - bu so'zlarni, talafotlarni, rasmlarni, tovushlarni yoki boshqalarni

discrete categorical datalarni
- low dimensional number vektoriga
o'tkazish.

Input (Hello) $\xrightarrow{\text{Transformerlar}}$ Neural Network $\Rightarrow$ output

Demak Embeddings foydalanuvchi
bigan inputni kompyuter tushu-
radigan shulga o'tkazib berish
bunda low dimensional vektorga
o'tadi va integer emas float bo'ladi

Embedding qilganda Distributed Representation o'lgan tushuncha bor
bu degani: xar bor sonda ma'lum bor
qim jareb bo'ladi xar bor raqam
and bor shisilga jareb bermaydi
Hello = [0.23, -0.48, 0.234, ...]

Semantic meaning - ma'no i'zhatidan
bor bo'lgan yagona so'zlar xisoblanadi

Embedding va Encoding farqi -
Embedding learned representation
yoni Sundy neyron tarmog' jamoatidan
organizatsion ifoda

Encoding esa Deterministic Conversion
sion - yoni qatay qoidalariga asos
langan konvertatsiya

Embedding. Inlari

Input - Model ichiga kiritiladi.

Output - Model hisoblab chiqarechi

Historical - eski usullar

Input Inlari $\begin{cases} \text{Token} \\ \text{RoPE} \end{cases}$

Output Inlari - Contextual

Historical Inlari $\begin{cases} \text{Static} \\ \text{Absolute pos} \\ \text{Relative pos} \end{cases}$

Position Embeddings

Absolute, Relative va RoPE
(Input Embeddings)

Meaning Based Embeddinglar

Static $\leftarrow$ $\rightarrow$ Contextual

Structure Based Embeddinglar

Token $\begin{cases} \text{Absolute pos} \\ \text{Relative pos} \end{cases} \rightarrow \text{RoPE (Rotary)}$

Local LLM da RoPE embedding
sundan foydalanamiz,

- Sanat standarti

- hatta usun sozni fushanish imkon
yatiga ega. 128.000 dan eshig
ssni fushuna oladi.

Attention

Sozlarga bo' - bo'ni bo'ishga imkon
berish va qaysi soz eng muhim
kanloga qaror qabul qilish imkon.

Eng - muhim - eng kuchli (muhim)
ssni topish emas pairwise relationships
anglashdir.

Foydalanishimiz: Input Embeddings
(Token + RoPE) $\rightarrow$ Transformer Layer
(in Attention)

Nima uchun kerak: No Attention No
Transformer No LLM (chatgpt)

Manosiz bog'lanmagan vektorlarni
(embeddingsdan kelgan) $[0.23, -0.22]$ -
manali bog'langan sozlarga aylanish
muchi mahallasi.

Attention tez, qisqa, kep xotimda
o'qiy iloh xususiyat/uga ega bo'lish
kerak.

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Q - query, K - key, V - value.

Attention ~~unlabeled~~

Architectural Types

MHA - multi head Attention

MQA - multi query Attention

GQA - Grouped-query Attention

Context Optimization Types

Sparse

Seeding

Window

LLM ~~unlabeled~~

Local LLM

Cloud Based LLM

Local LLM - own computerda, internet
beach emas, ownash seylash murash
safat shorcha (7B-400B paramet),
bechi GPU. 16 GB+ VRAM sahur.

Cloud Based LLM - cloud serverda,
dovos shloe beach, internet maybur,
safat yuzari, watola bargarar
resh, Claude Sonnet, Claude
opus 4.8 (Anthropic API / Claude AI
organi)